ANSWER KEY — Integration Lesson 19 Test: Initial Values & Net Change

For the grader · full working shown · give credit for correct method with small arithmetic slips · answers should satisfy BOTH given facts (derivative and point)

1. Given $f'(x) = 2x$ and $f(0) = 5$, find $f(x)$.

> **Work:** integrate: $f(x) = x^2 + C$. Plug in $x = 0$: $f(0) = 0 + C = 5$, so $C = 5$. Check: derivative of $x^2 + 5$ is $2x$ ✓, $f(0) = 5$ ✓. **Answer: $f(x) = x^2 + 5$.**

2. Given $f'(x) = 3x^2 + 2$ and $f(1) = 8$, find $f(x)$.

> **Work:** integrate term by term: $f(x) = x^3 + 2x + C$. Plug in $x = 1$: $1 + 2 + C = 3 + C = 8 \rightarrow C = 5$. Check: derivative of $x^3 + 2x + 5$ is $3x^2 + 2$ ✓, $f(1) = 1 + 2 + 5 = 8$ ✓. **Answer: $f(x) = x^3 + 2x + 5$.**

3. $v(t) = 8$ m/s, $s(0) = 2$ m. Find $s(t)$ and $s(3)$.

> **Work:** $s(t) = 8t + C$; $s(0) = C = 2$, so $s(t) = 8t + 2$. Then $s(3) = 24 + 2 = 26$. Sanity check: 8 m/s for 3 s = 24 m, starting from 2 m → 26 m ✓. **Answer: $s(t) = 8t + 2$; $s(3) = 26$ meters.**

4. $v(t) = 2t$ m/s, $s(2) = 9$ m. Find $s(t)$ and $s(4)$.

> **Work:** $s(t) = t^2 + C$. The known point is at $t = 2$ (NOT $t = 0$), so plug in $t = 2$: $4 + C = 9 \rightarrow C = 5$. So $s(t) = t^2 + 5$, and $s(4) = 16 + 5 = 21$. Check: $s(2) = 4 + 5 = 9$ ✓. **Answer: $s(t) = t^2 + 5$; $s(4) = 21$ meters.** (Common error: writing $C = 9$ by treating $s(2)$ like $s(0)$.)

5. Tank: 15 L at $t = 0$, fills at $r(t) = 2 + 4t$ L/min. Amount at $t = 3$?

> **Work:** net change theorem: amount $= 15 + \int_0^3 (2 + 4t)\,dt$. Antiderivative: $2t + 2t^2$ (check: derivative is $2 + 4t$ ✓). Evaluate: $(6 + 18) - 0 = 24$. Amount $= 15 + 24 = 39$. **Answer: 39 liters.**

6. Start with \$30, earn at $e(t) = 5 + 2t$ dollars/day. Money after 4 days?

> **Work:** earnings $= \int_0^4 (5 + 2t)\,dt = [5t + t^2]_0^4 = 20 + 16 = 36$ (check: derivative of $5t + t^2$ is $5 + 2t$ ✓). Balance $= 30 + 36 = 66$. **Answer: \$66.**

7. $F(2) = 7$ and $\int_2^5 F'(t)\,dt = 12$. Find $F(5)$ and explain.

> **Work:** net change theorem: $F(5) = F(2) + \int_2^5 F'(t)\,dt = 7 + 12 = 19$. Sentence: "the later amount equals the starting amount plus the accumulated change" — the integral of the rate over the in-between time IS the total change, so adding it to where you started gives where you end up. **Answer: $F(5) = 19$.**

8. $v(t) = 6t$, $s(0) = 4$. Find $s(2)$ two ways.

> **Work:** (a) Family: $s(t) = 3t^2 + C$ (check: derivative $6t$ ✓); $s(0) = C = 4 \rightarrow s(t) = 3t^2 + 4$; $s(2) = 12 + 4 = 16$. (b) $s(2) = s(0) + \int_0^2 6t\,dt = 4 + [3t^2]_0^2 = 4 + 12 = 16$. Both outfits agree. **Answer: $s(2) = 16$ meters both ways.**

9. $A(t) = 4t + C$ liters — what does C mean, and what would pin it down?

> **Work:** plug in $t = 0$: $A(0) = C$. So C is the amount of water in the tank at the start — the initial amount. The rate (4 L/min) only describes how the amount CHANGES; it cannot reveal how much was already there. One measurement of the tank at any single time (most simply, at $t = 0$) determines C. **Answer: C = the starting amount $A(0)$; knowing the amount at one moment pins it down.**

10. Plant: 5 cm at $t = 0$, grows at $g(t) = 2t + 2$ cm/week. (a) $h(t)$? (b) When 13 cm?

> **Work:** (a) family: $h(t) = t^2 + 2t + C$ (check: derivative is $2t + 2$ ✓); $h(0) = C = 5 \rightarrow h(t) = t^2 + 2t + 5$. (b) Set $h(t) = 13$: $t^2 + 2t + 5 = 13 \rightarrow t^2 + 2t - 8 = 0 \rightarrow (t + 4)(t - 2) = 0 \rightarrow t = -4$ (rejected, time can't be negative) or $t = 2$. Check: $h(2) = 4 + 4 + 5 = 13$ ✓. **Answer: (a) $h(t) = t^2 + 2t + 5$ (b) at week $t = 2$.**